

PDF ChatBot — RAG Project

An AI-powered chatbot that answers questions from any PDF using RAG (Retrieval-Augmented Generation).

Tech Stack

- Python
- LangChain
- FAISS (Vector Store)
- OpenAI API
- Streamlit (UI)

How to Run

Step 1 — Install dependencies

```
``bash
pip install -r requirements.txt
``
```

Step 2 — Run the app

```
``bash
streamlit run app.py
``
```

Step 3 — Use the app

1. Enter your OpenAI API Key
2. Upload any PDF
3. Ask questions!

Project Structure

```
rag-chatbot/
├── app.py          # Main application
├── requirements.txt # Dependencies
└── README.md       # This file
``
```

How it Works

1. PDF upload → extract text
2. Split text into chunks
3. Convert chunks to vectors (FAISS)

4. User asks question → find relevant chunks → AI answers

🎯 Resume Description

> Built a RAG pipeline that answers questions over uploaded PDFs using LangChain, FAISS vector store, and OpenAI API — served via a Streamlit web UI with source citation.